

Solution:

Introduction:

Phylogenetics' is the systematic study of reconstructing the past evolutionary history of extant species or taxa, based on present-day data, such as morphologies or molecular information (sequence data). This evolutionary history or phylogeny is ideally represented as a binary tree.

Phylogenetics for Biotechnologists:

Phylogenetics is important because it enriches our understanding of how genes, genomes, species (and molecular sequences more generally) evolve. Through phylogenetics, we learn not only how the sequences came to be the way they are today, but also general principles that enable us to predict how they will change in the future. This is not only of fundamental importance but also extremely useful for numerous applications for Biotechnologists:

1.Identifying the origin of pathogens:

Molecular sequencing technologies and phylogenetic approaches can be used to learn more about a new pathogen outbreak. This includes finding out about which species the pathogen is related to and subsequently the likely source of transmission. This can lead to new recommendations for public health policy.

2.Pharmalogical Significance:

Phylogenetic screening of pharmacologically related species can help identify closely related members of a species with pharmacological significance.

3.Phylogenetics in Microbiology:

Phylogenetic analysis explores the evolutionary relationships between organisms and is a vital foundation for microbial studies. The development of reliable phylogenetic trees is an important step in characterizing new pathogens and developing new treatments in biomedicine. Prokaryotic organisms evolved around 3 billion years ago, representing a time that included the majority of early evolutionary life history. Studies of prokaryotic evolutionary relationships are important for understanding the origin and diversification of life on earth.

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4.Proteine Structure and function Prediction:

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5.Medicinal Plants:

Plants are a dominant source of pharmacological drugs for the treatment and cure of different disorders and diseases. However, selecting the most biologically active plant species for further screening is still challenging. Phylogeny has strong explanatory powers and provides predictive perspectives that are not available in traditional plant classifications. China, which is endowed with a diverse set of therapeutic cures from Mother Nature, represents an ideal environment for the phylogenetic analysis of potential medicinal plants.

6.Role in Medicine Field.

The rapid growth rate of human population, along with the public health crisis encountered in many regions, particularly in developing world, create an urgent need for the discovery of alternative drugs. Because medicinal plants are not distributed randomly across lineages, it has been suggested that phylogeny along with traditional knowledge of plant uses can guide the identification of new medicinally useful plants. In this study, we combined different statistical approaches to test for phylogenetic signal in 33 categories of plant uses in South Africa. Depending on the null models considered, we found evidence for signal in up to 45% of plant use categories, indicating the need for multiple tests combination to maximize the chance of discovering new medicinal plants when applying a phylogenetic comparative approach. Furthermore, although there was no signal in the diversity of medicinal uses – i.e. total number of medicinal uses recorded for each plant – our results indicate that taxa that are evolutionarily closely related have significantly more uses than those that are evolutionarily isolated. Our study therefore provides additional support to the body of literature that advocates for the inclusion of phylogeny in bio screening medicinal flora for the discovery of alternative medicines.

Regards:

BT-201 Expert Zoologists: